

Lesson 03 Practice: Bits, Bytes & Units

JKSSB Computer Awareness

October 10, 2026

Provenance: 19 original JKSSB-pattern questions and one paraphrased PYQ adaptation from the official Website Operator 2026 booklet. The source item is not reproduced verbatim.

Q1. *[PYQ-adapted: JKSSB Website Operator 2026 Q72; official provisional answer key; paraphrased]* In the usual hierarchy of digital data units, what is the smallest unit?

- A. Block
- B. Bit
- C. Byte
- D. Word

Q2. *[Original]* How many bits are in one nibble?

- A. 2
- B. 4
- C. 8
- D. 16

Q3. *[Original]* How many bits are in one byte?

- A. 4
- B. 6
- C. 8
- D. 10

Q4. *[Original]* One byte is equal to:

- A. Two nibbles
- B. Two bits
- C. Four bytes
- D. Eight nibbles

Q5. *[Original]* What does lowercase **b** ordinarily denote in data-rate notation?

- A. Byte
- B. Bit
- C. Binary file

D. Buffer

Q6. *[Original]* Which notation is a rate rather than a storage amount?

- A. 20 MB
- B. 20 GB
- C. 20 Mbps
- D. 20 bytes

Q7. *[Original]* Under the lesson's 1024 convention, 1 KB is:

- A. 100 bytes
- B. 1000 bytes
- C. 1024 bytes
- D. 2048 bytes

Q8. *[Original]* How many bytes are in 3 KB under the 1024 convention?

- A. 3072
- B. 3000
- C. 24
- D. 1027

Q9. *[Original]* Under the same convention, 2 MB equals:

- A. 512 KB
- B. 1024 KB
- C. 2048 KB
- D. 2048 bytes

Q10. *[Original]* At matching decimal prefixes, 8 Mbps is equal to which ideal rate before protocol overhead?

- A. 1 MB/s
- B. 8 MB/s
- C. 64 MB/s
- D. 0.125 MB/s

Q11. *[Original]* A 64-bit processor word contains how many bytes?

- A. 4
- B. 8
- C. 16
- D. 64

Q12. *[Original]* Which statement about word size is correct?

- A. It is always exactly 8 bits.

- B. It is the same as a nibble on every processor.
- C. It depends on the processor architecture.
- D. It always means one character.

Q13. *[Original]* Which statement is **NOT** correct?

- A. Lowercase **b** commonly denotes bit.
- B. Uppercase **B** commonly denotes byte.
- C. A byte contains 8 bits.
- D. A word has one universal size on all processors.

Q14. *[Original]* A file is 4 MB. Which interpretation is correct?

- A. It is a storage amount measured in bytes.
- B. It is a data rate measured per second.
- C. It is 4 megabits per second.
- D. It specifies the processor word size.

Q15. *[Original]* Evaluate the statements:

- I. A nibble is 4 bits.
- II. A byte is 8 bits.
- III. A byte is two nibbles.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II, and III

Q16. *[Original]* A connection is rated at 24 Mbps. Ignoring overhead, what is its ideal byte rate using matching decimal prefixes?

- A. 3 MB/s
- B. 24 MB/s
- C. 192 MB/s
- D. 0.3 MB/s

Q17. *[Original]* Which statement correctly distinguishes capacity from transfer rate?

- A. Capacity is how much data can be held; rate is how much moves per time.
- B. Capacity is always measured per second; rate never includes time.
- C. Both terms mean processor word size.
- D. A larger capacity guarantees a faster transfer rate.

Q18. *[Original]* The 1024 ladder used in this lesson gives 1 GB as:

- A. 1024 MB

- B. 1000 MB
 - C. 8 MB
 - D. 1024 bytes
- Q19.** *[Original]* Why must a question's stated unit convention be checked before a capacity conversion?
- A. Decimal prefixes use powers of 1000 while the traditional binary ladder uses powers of 1024.
 - B. Bytes have different sizes in every file.
 - C. A bit is sometimes 4 bytes.
 - D. Capacity always depends on network speed.
- Q20.** *[Original]* A character is stored using UTF-8. Which assumption is unsafe?
- A. The encoded size may vary.
 - B. Every character must occupy exactly one byte.
 - C. Encoding represents text as bytes.
 - D. Character storage depends on the encoding.
- Q21.** *[JKSSB FAA-pattern]* How many bits make one byte?
- A. 8
 - B. 4
 - C. 16
 - D. 1024
- Q22.** *[JKPSI-pattern]* One kilobyte (KB) equals how many bytes?
- A. 1024
 - B. 100
 - C. 8
 - D. 1000 bits
- Q23.** *[SSC CGL-pattern]* Which of the following storage units is the largest?
- A. Terabyte
 - B. Gigabyte
 - C. Megabyte
 - D. Kilobyte
- Q24.** *[JKSSB-pattern]* (MSQ: select ALL correct options.) Which of the following conversions are correct?
- A. 4 bits = 1 nibble
 - B. 8 bits = 1 byte
 - C. 1024 bytes = 1 KB

D. 1000 KB = 1 MB

Q25. [SSC CGL-pattern] (MSQ: select ALL correct options.) Which statements about the bit and the byte are correct?

- A. The bit is the smallest unit of data.
- B. One byte consists of 8 bits.
- C. One nibble consists of 4 bits.
- D. A word always means exactly 8 bits on every machine.

Q26. [PYQ-adapted: JKSSB Website Operator 2026 Q72, provisional key; SRC-04/SRC-07] What is the smallest unit of data?

- A. Bit
- B. Byte
- C. Nibble
- D. Word

Q27. [Original] How many bits are in one byte?

- A. 4
- B. 8
- C. 16
- D. 32

Q28. [Original] How many nibbles make one byte?

- A. One
- B. Two
- C. Four
- D. Eight

Q29. [Original] Using the binary convention, how many bytes are in 1 KiB?

- A. 1000
- B. 1024
- C. 100
- D. 2048

Q30. [Original, MSQ drill] (MSQ: select ALL correct options.) Which statements are correct?

- A. A nibble contains four bits.
- B. A byte contains eight bits.
- C. A bit can represent 0 or 1.
- D. A byte contains four nibbles.